# AISC5020
# Week 6
# Information Theory: Entropy, Cross-Entropy and KL Divergence

Presented by

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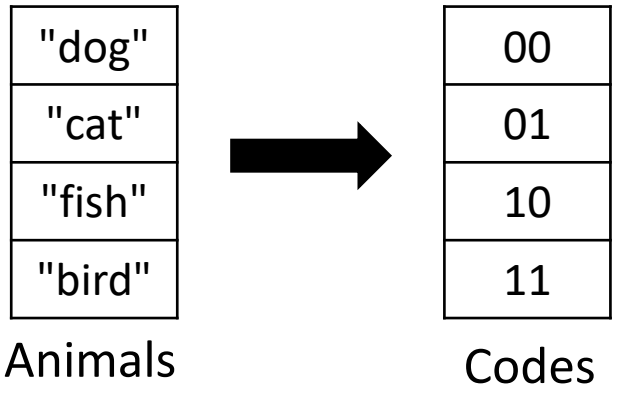
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# Chapter Outline

- Entropy, cross-entropy as loss
- KL divergence as distribution discrepancy
- Why CE = H + KL and why CE is used for classification

# Entropy

- Suppose that Bob says four words "dog", "cat", "fish" and "bird". Bob would like to communicate with you using the following codes.



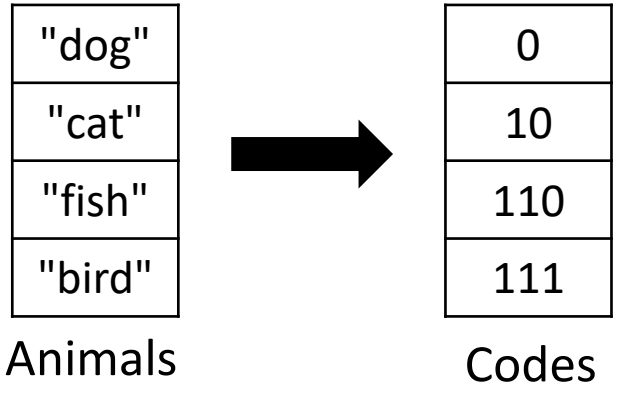

- To send a message, Bob replaces each animal with the corresponding code and then concatenate them together to form the encoded string.

| 00010011 | Encoded string |
|---|---|
| 00 01 00 11 | codes |
| dog cat dog bird | Source animals |

- Since each animal uses a 2-bit code, the average length of codes is 2 bits.

# Entropy

- Suppose that the relative frequencies Bob talks about the animals are "dog" 50%, "cat" 25%, "fish" 12.5% and "bird" 12.5%.

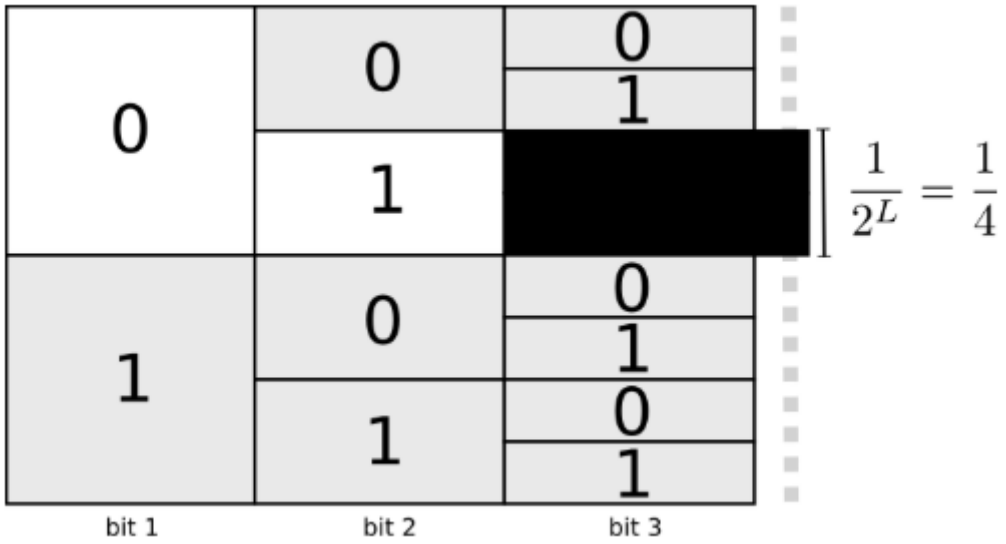



- By considering the above coding scheme, the average length of codes is

$$0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 + 0.125 \times 3 = 1.75 \text{ bits.}$$

- Let $p(x)$ be the probability (or relative frequency) of word (or code) $x$ and $L(x)$ be the number of bits (length) for code $x$. The average length of codes is $\sum_x p(x)L(x)$ bits.
- A code is said to have the prefix property if no code is a prefix of any other code. Example, "0" is not a prefix to "10", "110", or "111".

# Entropy




- Use of "0" (a code of 1 bit) as a code sacrifices $\frac{1}{2}$ of the possible codes for other codes.
- Use of a code of 2 bits length sacrifices $\frac{1}{4}$ of the possible codes for other codes.
- Use of a code of $L$ bits length sacrifices $\frac{1}{2^L}$ of the possible codes for other codes.
- Consider $\frac{1}{2^L}$ as the cost of using a code of length $L$. Since the use of a code of length $L = 0$ forbids us to use any code (sacrifices all possible codes), the total available cost is $1$.

# Entropy

- For each word $x$, we have a tuple $\left(p(x), L(x), \frac{1}{2^{L(x)}}\right)$. The average code length is $A = \sum_x p(x)L(x)$ and the total cost is $\sum_x \frac{1}{2^{L(x)}} \leq 1$.
- We want to find $L(x)$ that minimizes $A$ subject to the total cost constraint. Define the Lagrangian

$$\mathcal{L} = \sum_x p(x)L(x) + \lambda\left(\sum_x \frac{1}{2^{L(x)}} - 1\right)$$

$$\frac{\partial \mathcal{L}}{\partial L(x)} = p(x) + \lambda(-\ln 2)\frac{1}{2^{L(x)}}$$

- Setting $\frac{\partial \mathcal{L}}{\partial L(x)} = 0$, we have $\frac{1}{2^{L(x)}} \propto p(x) \Longrightarrow L(x) \propto \log_2\left(\frac{1}{p(x)}\right)$.
- Since $\sum_x p(x) = 1$, $L(x) = \log_2\left(\frac{1}{p(x)}\right)$ and $\sum_x \frac{1}{2^{L(x)}} = 1$ (Kraft's constraint).
- The optimal average code length is $H_p(p) = \sum_x p(x)\log_2\left(\frac{1}{p(x)}\right) = E_p\left(\log_2\left(\frac{1}{p(X)}\right)\right)$. We call this quantity the **entropy** of $p$. (Assume $0 \cdot \log_2 0 = 0$)

# Entropy

- Example. Consider a random variable $X$ that has a uniform distribution over 32 outcomes. The entropy of this random variable is

$$H(X) = \sum_{i=1}^{32} \frac{1}{32} \log_2 \left( \frac{1}{\frac{1}{32}} \right) = \log_2 2^5 = 5.$$

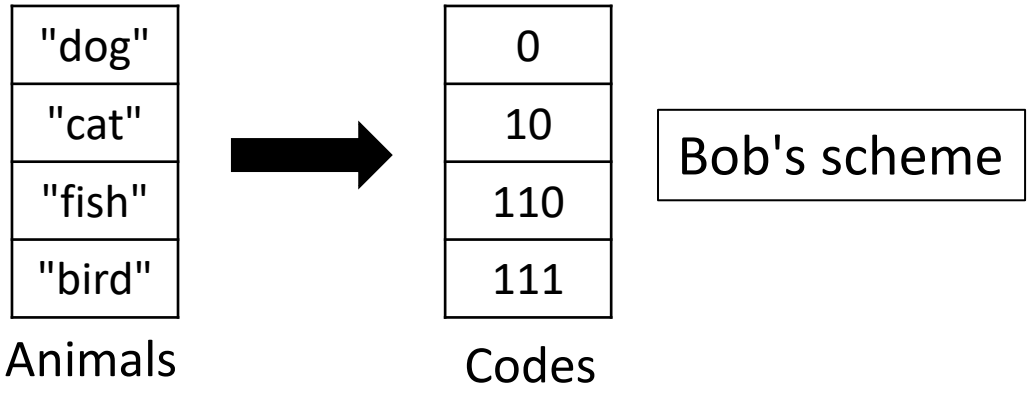
- The optimal average code length is 5 bits.

It measures the amount of uncertainty of the random variable $X$.

- Note that $H(X) \geq 0$ or $H_p(p) \geq 0$ since $\frac{1}{p(X)} \geq 1$.
- Since the cost associated with the use of a code $x$ of length $L(x)$ is $\frac{1}{2^{L(x)}} = p(x)$. The optimal average code length $A = \sum_x p(x)L(x)$ is the (minimum) total weighted cost (weighted by length of codes).
- Note that $H_p(p) \leq A < H_p(p) + 1$ if we require $L(x)$s are all integers.
  - By optimality construction, $H_p(p)$ minimizes $A$. Thus, $A \geq H_p(p)$.
  - If $L(x) = \left\lceil \log_2 \left( \frac{1}{p(x)} \right) \right\rceil$, $L(x) \geq \log_2 \left( \frac{1}{p(x)} \right) \Rightarrow 2^{-L(x)} \leq p(x) \Rightarrow \sum_x 2^{-L(x)} \leq \sum_x p(x) = 1$.
  - Moreover, $L(x) < \log_2 \left( \frac{1}{p(x)} \right) + 1 \Rightarrow A \leq H_p(p) + 1$.

# Cross Entropy

- Suppose Alice's relative frequencies of the animals are "dog" 12.5%, "cat" 50%, "fish" 25% and "bird" 12.5%.
- Currently Bob is using the following scheme to communicate animals.




- Using this scheme, Alice's average code length is $0.125 \times 1 + 0.5 \times 2 + 0.25 \times 3 + 0.125 \times 3 = 2.25$ bits. This average code length is called the **cross-entropy** of Alice's distribution with respect to Bob's distribution (Bob's scheme).

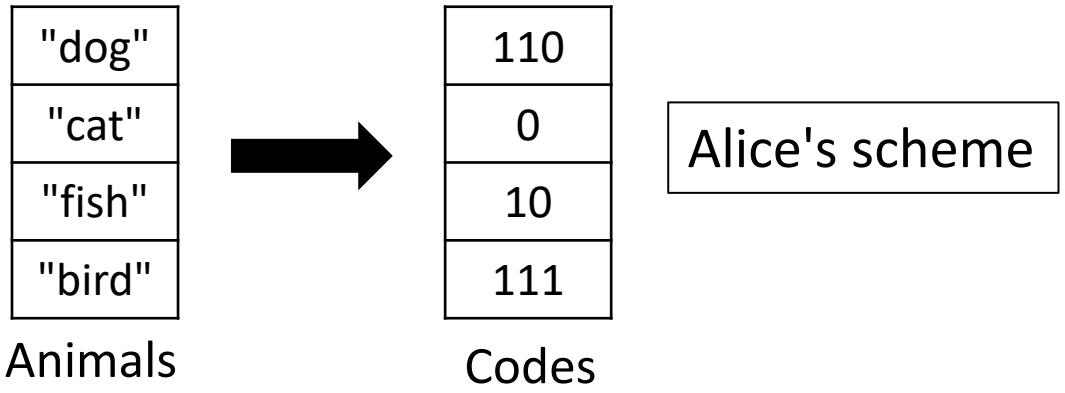
# Cross Entropy

- In general, the cross entropy of a distribution $q(x)$ with respect to another distribution $p(x)$ is

$$H_p(q) = \sum_x \underbrace{q(x)}_{\text{True Distribution}} L_p(x) = \sum_x q(x) \log_2 \left( \frac{1}{p(x)} \right) = E_q \left( \log_2 \left( \frac{1}{p(X)} \right) \right)$$

- Note that distributions $p$ and $q$ share the same set of codes (support).

- The entropy (optimum average code length) for Alice is $0.125 \times 3 + 0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 = 1.75$ bits. However, the average code length for Bob with respect to Alice's distribution is $0.5 \times 3 + 0.25 \times 1 + 0.125 \times 2 + 0.125 \times 3 = 2.375$ bits.




- In general, $H_p(q) \neq H_q(p)$.

# Cross Entropy

- $H_q(p) \geq H_p(p)$

$$H_q(p) - H_p(p) = \sum_x p(x) \log_2 \left(\frac{1}{q(x)}\right) - \sum_x p(x) \log_2 \left(\frac{1}{p(x)}\right) = \sum_x p(x) \log_2 \left(\frac{p(x)}{q(x)}\right) \geq 0.$$

- For $t \geq 1$, $\ln t \leq t - 1$ (the function $f(t) = \ln t - t + 1$ is a decreasing function on $[1, \infty)$).
- For $0 < t \leq 1$, $\ln t \leq t - 1$ (the function $f(t) = \ln t - t + 1$ is an increasing function on $(0,1]$).
- Let $t = \frac{q(x)}{p(x)}$. We have $\ln \frac{q(x)}{p(x)} \leq \frac{q(x)}{p(x)} - 1 \Longrightarrow -p(x) \ln \frac{q(x)}{p(x)} \geq -q(x) + p(x)$

$$\Longrightarrow \sum_x p(x) \ln \frac{p(x)}{q(x)} \geq \sum_x \big(p(x) - q(x)\big) = 0$$

- since $\sum_x p(x) = \sum_x q(x) = 1$. Also, because $\log_2 t = \frac{\ln t}{\ln 2}$, the result follows.

# Cross Entropy

- Since $H_q(p) - H_p(p) \geq 0$, we call the quantity $D_q(p) = H_q(p) - H_p(p) \geq 0$ the **Kullback-Leibler divergence**, or **KL divergence** of $p$ with respect to $q$.
- Literature also uses $D(p||q)$.

True Distribution

- Since $D_q(p) > 0$ if $q \neq p$ and $D_q(p) = 0$ if $q = p$, it behaves like a distance (but not a distance) between $p$ and $q$.
- In machine learning, we often use KL divergence to measure the distance between a true distribution $p$ (based on data) and a model or approximate or predicted distribution $q$.
- Since $H_q(p)$ is the cost of using $q$ for $p$ (Bob uses Alice's scheme), $D_q(p)$ is the extra cost due to using $q$ (but not $p$).

# Cross Entropy

- Example
- Suppose we have a coin with probability of getting a Head is $p = 0.9$ and we model the probability of head as $\hat{p} = 0.5$.

$$D(p||\hat{p}) = H_{\hat{p}}(p) - H_p(p)$$
$$= 0.9 \times \log_2\left(\frac{1}{0.5}\right) + 0.1 \times \log_2\left(\frac{1}{0.5}\right) - 0.9 \times \log_2\left(\frac{1}{0.9}\right) - 0.1 \times \log_2\left(\frac{1}{0.1}\right) = 1 - 0.469$$
$$= 0.531 \text{ bits}$$

- This is the extra cost due to using $\hat{p}$ but not $p$.

# Joint Entropy

- Consider the contingency table

| | T-shirt | Coat | Total |
|---|---|---|---|
| **Raining** | 0.06 | 0.19 | 0.25 |
| **Sunny** | 0.56 | 0.19 | 0.75 |
| **Total** | 0.62 | 0.38 | 1.00 |

- To communicate the weather and what I wear, we use the following codes

| | |
|---|---|
| "Sunny and T-shirt" | 0 |
| "Coat and Raining" | 10 |
| "Sunny and Coat" | 111 |
| "Raining and T-shirt" | 110 |

Codes

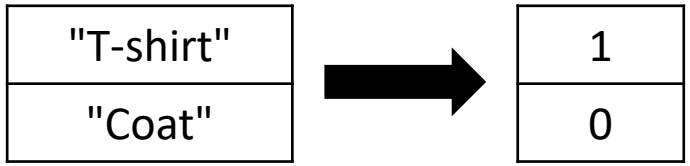
- The optimal average code length is $0.56 \times 1 + 0.19 \times 2 + 0.19 \times 3 + 0.06 \times 3 = 1.69$ bits.
- The joint entropy of two random variables $X$ and $Y$ is $H(X,Y) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{1}{p(x,y)}\right)$

# Joint Entropy

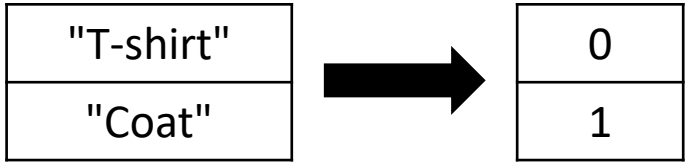
- Suppose we knows the weather (we can check it on the news).

| | T-shirt | Coat | Total |
|---|---|---|---|
| **Raining** | 0.06 | 0.19 | 0.25 |
| **Sunny** | 0.56 | 0.19 | 0.75 |
| **Total** | 0.62 | 0.38 | 1.00 |

- Given Raining, $P(\text{T-shirt} \mid \text{Raining}) = \frac{0.06}{0.25} = 0.24$ and $P(\text{Coat} \mid \text{Raining}) = \frac{0.19}{0.25} = 0.76$




- Given Sunny, $P(\text{T-shirt} \mid \text{Sunny}) = \frac{0.56}{0.75} = 0.747$ and $P(\text{Coat} \mid \text{Sunny}) = \frac{0.19}{0.75} = 0.253$



# Conditional Entropy

- The conditional entropy of $X$ given $Y$ is

$$H(X|Y) = \sum_y p(y) H(X|Y=y) = \sum_y p(y) \sum_x p(x|y) \log_2 \left( \frac{1}{p(x|y)} \right)$$

$$= \sum_{(x,y)} p(x,y) \log_2 \left( \frac{1}{p(x|y)} \right) = E_{X,Y} \left( \log_2 \left( \frac{1}{p(X|Y)} \right) \right)$$

- We can show that

$$H(X = \text{clothes}|Y = \text{weather}) = 0.25 H(X|Y = \text{Raining}) + 0.75 H(X|Y = \text{Sunny})$$

$$= 0.25 \left( 0.24 \times \log_2 \left( \frac{1}{0.24} \right) + 0.76 \times \log_2 \left( \frac{1}{0.76} \right) \right)$$

$$+ 0.75 \left( 0.747 \times \log_2 \left( \frac{1}{0.747} \right) + 0.253 \times \log_2 \left( \frac{1}{0.253} \right) \right) = 0.810759 \text{ bits}$$

# Conditional Entropy

- $H(X,Y) = H(Y) + H(X|Y)$

$$H(X,Y) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{1}{p(x,y)}\right) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{1}{p(x|y)p(y)}\right)$$

$$= \sum_{(x,y)} p(x,y) \log_2 \left(\frac{1}{p(y)}\right) + \sum_{(x,y)} p(x,y) \log_2 \left(\frac{1}{p(x|y)}\right) = \sum_{y} p(y) \log_2 \left(\frac{1}{p(y)}\right) + H(X|Y)$$

$$= H(Y) + H(X|Y)$$

# Conditional Entropy

- $H(X,Y) \geq H(X) \geq H(X|Y)$
- Since $H(X,Y) = H(X) + H(Y|X)$ and $H(Y|X) \geq 0$, $H(X,Y) \geq H(X)$
- Since $D_{p(X)p(Y)}\big(p(X,Y)\big) \geq 0$,

$$\sum_{(x,y)} p(x,y) \log_2 \left( \frac{p(x,y)}{p(x)p(y)} \right) \geq 0$$

$$\Rightarrow \sum_{(x,y)} p(x,y) \log_2 p(x,y) - \sum_{(x,y)} p(x,y) \log_2 p(x) - \sum_{(x,y)} p(x,y) \log_2 p(y) \geq 0$$

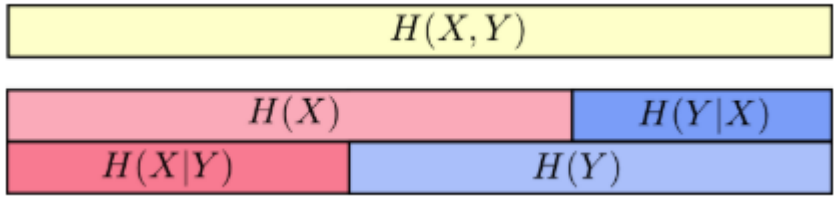
$$\Rightarrow -H(X,Y) + H(X) + H(Y) \geq 0$$

$$\Rightarrow \boxed{H(X,Y) \leq H(X) + H(Y)}$$

$$\Rightarrow H(X|Y) = H(X,Y) - H(Y) \leq H(X)$$

# Conditional Entropy

- $H(X,Y) \geq H(X) \geq H(X|Y)$ since $H(X,Y) = H(X) + H(Y|X)$ and $H(Y|X) \geq 0$
- Adding variables increases uncertainty, while conditioning reduces it.
- $H(X,Y|Z) = H(Y|Z) + H(X|Y,Z)$
- $H(X|Y) \neq H(Y|X)$
- $H(X) - H(X|Y) = H(Y) - H(Y|X)$ since $H(X,Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$



# Mutual Information

- Since $H(X,Y) \leq H(X) + H(Y)$, define $I(X,Y) = H(X) + H(Y) - H(X,Y)$ as the **mutual information** between $X$ and $Y$.
- $I(X,Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$

mutual information as how much knowing $Y$ reduces your uncertainty about $X$.

- $I(X,Y) \geq 0$; $I(X,Y) = 0$ if and only if $X$ and $Y$ are independent.
- $I(X,Y|Z) \geq 0$; $I(X,Y|Z) = 0$ if and only if $X$ and $Y$ are conditional independent given $Z$.
- $H(X,Y) - I(X,Y) = H(X|Y) + H(Y|X)$

$$H(X,Y) - I(X,Y) = H(X,Y) - \big(H(X) + H(Y) - H(X,Y)\big)$$
$$= \big(H(X,Y) - H(X)\big) + \big(H(X,Y) - H(Y)\big) = H(X|Y) + H(Y|X)$$

- $I(X,Y) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(x,y)}{p(x)p(y)}\right) = E_{X,Y}\left(\log_2 \left(\frac{p(X,Y)}{p(X)p(Y)}\right)\right) = D_{p(X)p(Y)}\big(p(X,Y)\big)$

# Mutual Information

- The variation of information between $X$ and $Y$ is defined as $V(X,Y) = H(X,Y) - I(X,Y) \geq 0$.
  - It is zero when knowing the value of one tells you the value of the other and increases as they become independent.
  - It is a distance measure between two jointly distributed variables.
  - KL divergence gives us a distance between two distributions over the same variable or set of variables.
- Example. Consider two binary random variables $X$ and $Y$ with the joint distribution:

| | $\boldsymbol{Y=0}$ | $\boldsymbol{Y=1}$ | |
|---|---|---|---|
| $\boldsymbol{X=0}$ | 0.4 | 0.1 | 0.5 |
| $\boldsymbol{X=1}$ | 0.1 | 0.4 | 0.5 |
| | 0.5 | 0.5 | |

- $H(X) = H(Y) = 0.5 \log_2\left(\frac{1}{0.5}\right) + 0.5 \log_2\left(\frac{1}{0.5}\right) = 1$ bit
- $H(X,Y) = 0.4 \log_2\left(\frac{1}{0.4}\right) + 0.1 \log_2\left(\frac{1}{0.1}\right) + 0.1 \log_2\left(\frac{1}{0.1}\right) + 0.4 \log_2\left(\frac{1}{0.4}\right) = 1.722$ bits
- $I(X,Y) = H(X) + H(Y) - H(X,Y) = 0.278$ bits
- $V(X,Y) = H(X,Y) - I(X,Y) = 1.444$ bits $\Longrightarrow X$ and $Y$ are not perfectly dependent.

# Chain Rules

- Chain Rule. Let $X_1, X_2, \dots, X_n$ be drawn according to $p(x_1, x_2, \dots, x_n)$. Then

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^{n} H(X_i | X_{i-1}, \dots, X_1)$$
$$= H(X_1) + H(X_2|X_1) + H(X_3|X_1, X_2) + \cdots + H(X_n|X_{n-1}, \dots, X_1)$$

- Proof.

$$H(X_1, X_2) = H(X_1) + H(X_2|X_1)$$
$$H(X_1, X_2, X_3) = H(X_1, X_2) + H(X_3|X_1, X_2)$$

- Assume $H(X_1, X_2, \dots, X_k) = \sum_{j=1}^{k} H\left(X_j \middle| X_{j-1}, \dots, X_1\right)$

$$H(X_1, X_2, \dots, X_{k+1}) = H(X_1, X_2, \dots, X_k) + H(X_{k+1}|X_k, \dots, X_1)$$
$$= \sum_{j=1}^{k} H\left(X_j \middle| X_{j-1}, \dots, X_1\right) + H(X_{k+1}|X_k, \dots, X_1)$$

# Chain Rules

- The conditional mutual information of $X$ and $Y$ given $Z$ is

$$I(X,Y|Z) = H(X|Z) - H(X|Y,Z)$$

- analogous to $I(X,Y) = H(X) - H(X|Y)$.

- Chain Rule for Information

$$I(X_1, X_2, \dots, X_n, Y) = \sum_{i=1}^{n} I(X_i, Y|X_{i-1}, \dots, X_1)$$

- Proof

$$I(X_1, X_2, \dots, X_n, Y) = H(X_1, X_2, \dots, X_n) - H(X_1, X_2, \dots, X_n|Y)$$

$$= \sum_{i=1}^{n} H(X_i|X_{i-1}, \dots, X_1) - \sum_{i=1}^{n} H(X_i|X_{i-1}, \dots, X_1, Y) = \sum_{i=1}^{n} I(X_i, Y|X_{i-1}, \dots, X_1)$$

# Chain Rules

- The conditional relative entropy

$$D_{q(Y|X)}\big(p(Y|X)\big) = \sum_x p(x) D_{q(Y|X=x)}\big(p(Y|X=x)\big) = \sum_x p(x) \sum_y p(y|x) \log_2 \left(\frac{p(y|x)}{q(y|x)}\right)$$

$$= \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(y|x)}{q(y|x)}\right)$$

- Chain Rule $D_{q(X,Y)}\big(p(X,Y)\big) = D_{q(X)}\big(p(X)\big) + D_{q(Y|X)}\big(p(Y|X)\big)$

$$D_{q(X,Y)}\big(p(X,Y)\big) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(x,y)}{q(x,y)}\right) = \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(x)p(y|x)}{q(x)q(y|x)}\right)$$

$$= \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(x)}{q(x)}\right) + \sum_{(x,y)} p(x,y) \log_2 \left(\frac{p(y|x)}{q(y|x)}\right) = D_{q(X)}\big(p(X)\big) + D_{q(Y|X)}\big(p(Y|X)\big)$$

# Markov Chain Inequality

- Random variables $X, Y, Z$ are said to form a Markov chain in that order, denoted by $X \to Y \to Z$, if the conditional distribution of $Z$ depends only on $Y$ and is conditionally independent of $X$. Specifically, $X, Y$ and $Z$ form a Markov chain $X \to Y \to Z$ if the joint probability mass function is factorized into $p(x, y, z) = p(x)p(y|x)p(z|y)$.

- Properties
  - $X \to Y \to Z$ if and only if $X$ and $Z$ are conditionally independent given $Y$.

$$p(x, z|y) = \frac{p(x, y, z)}{p(y)} = \frac{p(x, y)p(z|y)}{p(y)} = p(x|y)p(z|y)$$

  - This characterization of Markov chains can be extended to define Markov fields, which are $n$ dimensional random processes in which the interior and exterior are independent given the values on the boundary.
  - $X \to Y \to Z$ implies that $Z \to Y \to X$. Thus, we can write $X \leftrightarrow Y \leftrightarrow Z$.

$$p(z)p(y|z)p(x|y) = p(z)p(y|z)p(x|y, z) = p(x, y, z)$$

# Markov Chain Inequality

- Properties
  - If $Z = f(Y)$, then $X \to Y \to Z$.

- Since $Z$ is a function of $Y$, $p(z|y) = \begin{cases} 1 & \text{if } z = f(y) \\ 0 & \text{otherwise} \end{cases}$.

$$p(x, y, z) = p(x)p(y|x)p(z|x, y) = p(x)p(y|x)p(z|y) = p(x)p(y|x)I\big(z = f(y)\big)$$

- If $X \to Y \to Z$, $I(X, Y) \geq I(X, Z)$
  - By the chain rule of mutual information (treating $X$ as $Y$ in the formula),

$$I(X, Y, Z) = I(X, Z) + I(X, Y|Z) = I(X, Y) + I(X, Z|Y)$$

  - The fact that $X$ and $Z$ are independent given $Y$ implies that $I(X, Z|Y) = 0$.
  - Since $I(X, Y|Z) \geq 0$, $I(X, Y) \geq I(X, Z)$.
- Since $X \to Y \to Z \Longrightarrow Z \to Y \to X$, we also have $I(Y, Z) \geq I(X, Z)$.
- If $Z = g(Y)$, then $X \to Y \to Z$ and we have $I(X, Y) \geq I\big(X, g(Y)\big)$.

# Markov Chain Inequality

- If $X \to Y \to Z$, $I(X,Y|Z) \leq I(X,Y)$
  - Since $I(X,Z|Y) = 0$ and $I(X,Z) + I(X,Y|Z) = I(X,Y) + I(X,Z|Y)$ and $I(X,Z) \geq 0$, we have $I(X,Y|Z) \leq I(X,Y)$.
  - The mutual information (dependence) between $X$ and $Y$ is reduced by the observation of $Z$.
  - If $X, Y$ and $Z$ do not form a Markov chain, it can happen that $I(X,Y|Z) > I(X,Y)$. For example, let $X$ and $Y$ be independent fair binary random variables, and let $Z = X + Y$.
  - Since $X = \begin{cases} 0, p = 0.5 \\ 1, p = 0.5 \end{cases}$ and $Y = \begin{cases} 0, p = 0.5 \\ 1, p = 0.5 \end{cases}$, $Z = \begin{cases} 0, p = 0.25 \\ 1, p = 0.50 \\ 2, p = 0.25 \end{cases}$.
  - When $Z = 0, X = 0$ with probability 1. So, $H(X|Z = 0) = 0$.
  - When $Z = 2, X = 1$ with probability 1. So, $H(X|Z = 2) = 0$.
  - When $Z = 1, X = 1$ with probability 0.5 and $X = 0$ with probability 0.5. So, $H(X|Z = 1) = 1$ bit.

# Markov Chain Inequality

- If $X \to Y \to Z$, $I(X,Y|Z) \leq I(X,Y)$
  - If $X, Y$ and $Z$ do not form a Markov chain, it can happen that $I(X,Y|Z) > I(X,Y)$. For example, let $X$ and $Y$ be independent fair binary random variables, and let $Z = X + Y$.
  - Since $X = \begin{cases} 0, p = 0.5 \\ 1, p = 0.5 \end{cases}$ and $Y = \begin{cases} 0, p = 0.5 \\ 1, p = 0.5 \end{cases}$, $Z = \begin{cases} 0, p = 0.25 \\ 1, p = 0.50 \\ 2, p = 0.25 \end{cases}$.
  - $I(X,Y) = 0$ but $I(X,Y|Z) = H(X|Z) - H(X|Y,Z) = H(X|Z) = \sum_z p(z) H(X|Z = z) = \frac{1}{2}$.

In general,

$$I(X;Y) = H(X) - H(X \mid Y),$$

while

$$I(X;Y \mid Z) = H(X \mid Z) - H(X|Y,Z).$$

Conditioning on $Z$ reduces **both** entropy terms. Which one decreases more determines whether conditional mutual information goes up or down. Conditioning always reduces uncertainty (entropy), but it can either weaken or strengthen the apparent relationship (mutual information) between two variables.

# Application on Inference

- Suppose that we have a family of probability functions $\{f_\theta(x)\}$ indexed by $\theta$, and let $X$ be a sample from a distribution in this family.
- Let $T(X)$ be any statistic (function of the sample) such as sample mean or sample variance.
- Then $\theta \to X \to T(X)$ and, hence, $I\big(\theta, T(X)\big) \leq I(\theta, X)$. This is true for any distribution on $\theta$.
- A function $T(X)$ is said to be a sufficient statistic relative to the family $\{f_\theta(x)\}$ if $X$ is independent of $\theta$ given $T(X)$ for any distribution on $\theta$, that is $\theta \to T(X) \to X$.
- In this case, $I\big(\theta, T(X)\big) = I(\theta, X)$ for any distributions on $\theta$.

# Application on Inference

- Example. Let $X_1, X_2, \ldots, X_n$, $X_i \in \{0,1\}$, be an independently and identically distributed (IID) sequence of coin tosses of a coin with unknown parameter $\theta = P(X_i = 1)$. Given $n$, the number of 1's is a sufficient statistic for $\theta$, that is $T(X_1, X_2, \ldots, X_n) = \sum_{j=1}^{n} X_j$. We can show that, given $T$, all sequences having that many 1's are equally likely and independent of $\theta$.
- $P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) = \theta^{T(x_1,x_2,\ldots,x_n)}(1-\theta)^{n-T(x_1,x_2,\ldots,x_n)}$
- $P(T = k) = \binom{n}{k}\theta^k(1-\theta)^{n-k}$
- $P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n | T = k) = \frac{\theta^k(1-\theta)^{n-k}}{\binom{n}{k}\theta^k(1-\theta)^{n-k}} = \frac{1}{\binom{n}{k}}$.

# Application on Inference

- Example. If $X$ is normally distributed with mean $\theta$ and variance 1, that is $f_\theta(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\theta)^2}{2}}$ and $X_1, X_2, \ldots, X_n$ are drawn independently according to this distribution, a sufficient statistic for $\theta$ is the sample mean $\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j$. It can be verified that the conditional distribution of $X_1, X_2, \ldots, X_n$ conditioned on $\bar{X}_n$ and $n$ does not depend on $\theta$.

$$f_\theta(x_1, x_2, \ldots, x_n) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_j-\theta)^2}{2}} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{j=1}^{n}(x_j-\theta)^2} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\left\{\sum_{j=1}^{n} x_j^2 - 2n\theta\bar{x}_n + n\theta^2\right\}}$$

$$= (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{j=1}^{n} x_j^2} e^{n\theta\bar{x}_n - \frac{1}{2}n\theta^2} = (2\pi)^{-\frac{n}{2}} e^{-\frac{1}{2}\sum_{j=1}^{n} x_j^2} e^{\frac{n}{2}\bar{x}_n^2} e^{-\frac{n}{2}(\bar{x}_n-\theta)^2}$$

- Since $\bar{X}_n \sim n\left(\theta, \frac{1}{n}\right)$, $g_\theta(t) = \sqrt{\frac{n}{2\pi}} e^{-\frac{n}{2}(t-\theta)^2}$.

- $h_\theta(x_1, x_2, \ldots, x_n | \bar{X}_n = t) = \frac{f_\theta(x_1, x_2, \ldots, x_n)}{g_\theta(t)} = \frac{1}{\sqrt{n}}(2\pi)^{-\frac{n-1}{2}} e^{-\frac{1}{2}\sum_{j=1}^{n} x_j^2} e^{\frac{n}{2}t^2} I(\bar{x}_n = t)$.

# Application on Inference

- Example. If $f_\theta = \text{Uniform}(\theta, \theta+1)$, a sufficient statistic for $\theta$ is

$$T(X_1, X_2, \dots, X_n) = (\max(X_1, X_2, \dots, X_n), \min(X_1, X_2, \dots, X_n))$$

- Note that $f_\theta(x) = I(\theta < x < \theta + 1)$

$$f_\theta(x_1, x_2, \dots, x_n) = \prod_{j=1}^{n} I\big(\theta < x_j < \theta + 1\big)$$
$$= I(\max(x_1, x_2, \dots, x_n) < \theta + 1) I(\min(x_1, x_2, \dots, x_n) > \theta)$$
$$= I(\max(x_1, x_2, \dots, x_n) - 1 < \theta < \min(x_1, x_2, \dots, x_n)) = g_\theta(T = t)$$

- It shows that $f_\theta(x_1, x_2, \dots, x_n | T = (u, v)) \propto \frac{f_\theta(x_1, x_2, \dots, x_n)}{g_\theta(T=t)} = 1$ does not depend on $\theta$.

$$f_\theta\big(T = t = (u, v)\big) = n(n-1)(u - v)^{n-2} I(u - 1 < \theta < v)$$

# Application on Inference

- If $H(X|Y) = 0$ then $X$ is a function of $Y$.
  - $H(X|Y) = \sum_y p(y)H(X|Y = y) = 0$. Since $p(y) \geq 0$ and $H(X|Y = y) \geq 0$, $H(X|Y) = 0$ can happen only when $H(X|Y = y) = 0$ for all $y$ with $p(y) > 0$.
  - A distribution has entropy $0$ if and only if it is deterministic, that is there is only one $x_0$ such that $p(X = x_0|Y = y) = 1$, and $p(X = x|y) = 0$ for other values of $x \neq x_0$.
  - For each $y$ with $p(y) > 0$, $H(X|Y = y) = 0$ implies that there is a unique $x_y = f(y)$ such that $p\big(X = x_y\big|Y = y\big) = 1$ and all other values $p(X = x|Y = y) = 0$.
  - This defines a function $f$ (on the support of $Y = \{y: p(y) > 0\}$) such that $X = f(Y)$, with probability 1.

# Application on Inference

- To estimate $X$ with low probability of error only if $H(X|Y)$ is small.
- To estimate $X$ with a distribution $p(x)$. We observe a random variable $Y$ that is related to $X$ by $p(y|x)$. From $Y$, we form an estimator of $X$ as $\hat{X} = g(Y)$. Note that the range of values $\hat{X}$ assumes may be different from that of $X$. With this construction, $X \to Y \to \hat{X}$ and the probability of error is $P_e = P(\hat{X} \neq X)$.

# Application on Inference

Example: Let $X$ be a transmitted bit $P(X=0)=P(X=1)=0.5$

- We observe $Y$ through a noisy channel $P(Y=x|X=x)=0.9$ and $P(Y\neq x|X=x)=0.1$
- Estimator $\hat{X}=g(Y)=Y$ (simple guess whatever we have received)
- Probability of error $P_e=P(\hat{X}\neq X)=0.1$
- The joint p.m.f.:

| | $Y=0$ | $Y=1$ |
|---|---|---|
| $X=0$ | 0.45 | 0.05 |
| $X=1$ | 0.05 | 0.45 |

$$P(X=0,Y=0)=P(Y=0|X=0)P(X=0)=0.9\times 0.5=0.45$$
$$P(X=0,Y=1)=P(Y=1|X=0)P(X=0)=0.1\times 0.5=0.05$$
$$P(X=1,Y=0)=P(Y=0|X=1)P(X=1)=0.1\times 0.5=0.05$$
$$P(X=1,Y=1)=P(Y=1|X=1)P(X=1)=0.9\times 0.5=0.45$$

- Conditional Entropy

$$H(X|Y)=H(X,Y)-H(Y)=2\left[0.45\log_2\left(\frac{1}{0.45}\right)+0.05\log_2\left(\frac{1}{0.05}\right)\right]-2\times 0.5\log_2\left(\frac{1}{0.5}\right)$$
$$=0.468 \text{ bits}$$

# Application on Inference

Example:

- Suppose we want to predict whether a student passes an exam $X = 1$ (pass) and $X = 0$ (fail), with $P(X = 1) = 0.7$.
- We observe $Y =$ hours studied (discretized: low/medium/high).
- Suppose we have the conditional probabilities $p(y|x)$:

| | $\boldsymbol{Y}$ = **Low** | $\boldsymbol{Y}$ = **Medium** | $\boldsymbol{Y}$ = **High** |
|---|---|---|---|
| $\boldsymbol{X} = \mathbf{0}$ (fail) | 0.6 | 0.3 | 0.1 |
| $\boldsymbol{X} = \mathbf{1}$ (pass) | 0.2 | 0.4 | 0.4 |

- A good estimator $\hat{X} = 1$ if $Y$ is medium or high and $\hat{X} = 0$ if $Y$ is low.
- Joint distribution:

$$P(X = 0, Y = Low) = P(Y = Low|X = 0)P(X = 0) = 0.6 \times 0.3 = 0.18$$
$$P(X = 0, Y = Medium) = P(Y = Medium|X = 0)P(X = 0) = 0.3 \times 0.3 = 0.09$$
$$P(X = 0, Y = High) = P(Y = High|X = 0)P(X = 0) = 0.1 \times 0.3 = 0.03$$

# Application on Inference

Example:

- Joint distribution:

$$P(X=0,Y=Low)=P(Y=Low|X=0)P(X=0)=0.6\times 0.3=0.18$$
$$P(X=0,Y=Medium)=P(Y=Medium|X=0)P(X=0)=0.3\times 0.3=0.09$$
$$P(X=0,Y=High)=P(Y=High|X=0)P(X=0)=0.1\times 0.3=0.03$$
$$P(X=1,Y=Low)=P(Y=Low|X=1)P(X=1)=0.2\times 0.7=0.14$$
$$P(X=1,Y=Medium)=P(Y=Medium|X=1)P(X=1)=0.4\times 0.7=0.28$$
$$P(X=1,Y=High)=P(Y=High|X=1)P(X=1)=0.4\times 0.7=0.28$$

| | $\boldsymbol{Y}=$ **Low** | $\boldsymbol{Y}=$ **Medium** | $\boldsymbol{Y}=$ **High** |
|---|---|---|---|
| $\boldsymbol{X}=\mathbf{0}$ (fail) | 0.18 | 0.09 | 0.03 |
| $\boldsymbol{X}=\mathbf{1}$ (pass) | 0.14 | 0.28 | 0.28 |

- $H(X|Y)=0.32H(X|Y=low)+0.37H(X|Y=Medium)+0.31H(X|Y=High)=$
  $0.32\left(\frac{0.18}{0.32}\log_2\left(\frac{0.32}{0.18}\right)+\frac{0.14}{0.32}\log_2\left(\frac{0.32}{0.14}\right)\right)+0.37\left(\frac{0.09}{0.37}\log_2\left(\frac{0.37}{0.09}\right)+\frac{0.28}{0.37}\log_2\left(\frac{0.37}{0.28}\right)\right)+$
  $0.31\left(\frac{0.03}{0.31}\log_2\left(\frac{0.31}{0.03}\right)+\frac{0.28}{0.31}\log_2\left(\frac{0.31}{0.28}\right)\right)=0.32\times 0.9887+0.37\times 0.8004+0.31\times 0.4587=0.7547$ bits
- $H(X|Y)$ is moderate and there would be some $P_e>0$ but reasonably small.

# Application on Inference

Example:

| | $\boldsymbol{Y = }$ **Low** | $\boldsymbol{Y = }$ **Medium** | $\boldsymbol{Y = }$ **High** |
|---|---|---|---|
| $\boldsymbol{X = 0}$ (fail) | 0.18 | 0.09 | 0.03 |
| $\boldsymbol{X = 1}$ (pass) | 0.14 | 0.28 | 0.28 |

An error occurs in two situations:

- $X = 1$ but $Y =$ Low
- $X = 0$ but $Y =$ Medium or High

Therefore,

$$P_e = P(X = 1, Y = Low) + P(X = 0, Y = Medium) + P(X = 0, Y = High) = 0.14 + 0.09 + 0.03 = 0.26$$

# Fano Inequality

- Assume that $X$ is a random variable with finite support $\mathcal{X}$, that is the range of values of $X$ such that $p(x) > 0$ is finite. For any estimator $\hat{X}$ such that $X \to Y \to \hat{X}$, with $P_e = P\left(\hat{X} \neq X\right)$, we have

$$H(P_e) + P_e \log_2 |\mathcal{X}| \geq H\left(X\middle|\hat{X}\right) \geq H(X|Y).$$

  where $H(P_e) = -P_e \log_2 P_e - (1 - P_e) \log_2 (1 - P_e)$.

- This inequality can be weakened to $1 + P_e \log_2 |\mathcal{X}| \geq H(X|Y)$ or $P_e \geq \dfrac{H(X|Y) - 1}{\log_2 |\mathcal{X}|}$.
- Note that $P_e = 0$ implies that $H(X|Y) = 0$.
- $H\left(X\middle|\hat{X}\right)$ is the remaining uncertainty about $X$ after seeing the estimate $\hat{X}$.
- Fano inequality gives a fundamental lower bound on $P_e$ or an upper bound for $H(X|Y)$.

# Fano Inequality

- Practical implication of Fano Inequality
  - If $H(X|Y) > 0$, no estimator can perfectly recover $X$.
  - There is a minimum unavoidable error rate, no matter how clever your method is.
  - $X$: true label, $Y$: observed features, $\hat{X}$: predicted values. If $H(X|Y)$ is large, $P_e$ must be large.
  - If your features don't contain enough information about the label, no model can perform well.
- Example. Suppose $|\mathcal{X}| = 8$ and $H(X|Y) = 3$ bits.

$$P_e \geq \frac{H(X|Y) - 1}{\log_2 |\mathcal{X}|} = \frac{3 - 1}{\log_2 8} = \frac{2}{3}$$

$\Longrightarrow$ We cannot achieve better than $\approx 66.67\%$ error rate.

# Fano Inequality

- Define $E = \begin{cases} 1 \text{ if } X \neq \hat{X} \\ 0 \text{ if } X = \hat{X} \end{cases}$. So, $P(E=1) = P_e$ and $P(E=0) = 1 - P_e$.
- Since $H(E, X|\hat{X}) = H(X|\hat{X}) + H(E|X, \hat{X})$ and $H(E|X, \hat{X}) = 0$, Analogous to $H(X,Y) = H(X) + H(Y|X)$

$$H(X|\hat{X}) = H(E, X|\hat{X}) = H(E|\hat{X}) + H(X|E, \hat{X})$$

- $H(E|\hat{X}) \leq H(E) = H(P_e)$
- $H(X|E, \hat{X}) = P_e H(X|E=1, \hat{X}) + (1 - P_e) H(X|E=0, \hat{X})$
  - Since $E = 0$, $X = \hat{X}$ and $H(X|E=0, \hat{X}) = 0$.
  - For $E = 1$, we only know $X \neq \hat{X}$. Since $X$ can be one of the values in $\mathcal{X}$, we must have $H(X|E=1, \hat{X}) \leq \log_2|\mathcal{X}|$ (see next page).
- Therefore, $H(X|\hat{X}) \leq H(P_e) + P_e \log_2|\mathcal{X}|$.

# Fano Inequality

- For $E = 1$, we only know $X \neq \hat{X}$. Since $X$ can be one of the values in $\mathcal{X}$ and $X \neq \hat{X}$, the total number of possible values $X$ can take is $|\mathcal{X}| - 1$.
- Since entropy measures uncertainty, what distribution over these $|\mathcal{X}| - 1$ values yields the maximum possible uncertainty?
- The answer is the uniform distribution. If $X$ is equally likely to be any of the $|\mathcal{X}| - 1$ incorrect values, our uncertainty is maximized.
- The entropy of a uniform distribution over $N$ items is $\sum_{j=1}^{N} \frac{1}{N} \log_2(N) = \log_2(N)$.
- Therefore, $H\left(X \middle| E = 1, \hat{X}\right) \leq \log_2(|\mathcal{X}| - 1) \leq \log_2|\mathcal{X}|$.

# Fano Inequality

- Note that $H(P_e) = -P_e \log_2 P_e - (1 - P_e) \log_2 (1 - P_e)$
- We can show that $H(P_e)$ achieves its maximum at $P_e = 0.5$ and $H(0.5) = 1$.
- Hence, $0 \leq H(P_e) \leq 1$ and $H\left(X \middle| \hat{X}\right) \leq H(P_e) + P_e \log_2 |\mathcal{X}| \leq 1 + P_e \log_2 |\mathcal{X}|$.

- For any two random variables $X$ and $Y$ with finite support $\mathcal{X}$, let $p = P(X \neq Y)$. We have

$$H(X|Y) \leq H(p) + p \log_2 |\mathcal{X}| .$$

# Fano Inequality

- Suppose that there is no knowledge of $Y$. Then, $X$ must be guessed without any information.
- Let $X \in \{1,2,\dots,m\}$ and $p_1 \geq p_2 \geq \cdots \geq p_m$. The best guess of $X$ is $\hat{X} = 1$ and $P_e = 1 - p_1$.
- The Fano inequality becomes $H(X) \leq H(P_e) + P_e \log_2(m-1)$.
- The probability mass function $(p_1, p_2, \dots, p_m) = \left(1 - P_e, \frac{P_e}{m-1}, \dots, \frac{P_e}{m-1}\right)$ achieves this bound with equality.

$$H(X) = -(1-P_e)\log_2(1-P_e) - (m-1)\frac{P_e}{m-1}\log_2\left(\frac{P_e}{m-1}\right)$$
$$= -(1-P_e)\log_2(1-P_e) - P_e \log_2 P_e + P_e \log_2(m-1)$$
$$H(P_e) = -(1-P_e)\log_2(1-P_e) - P_e \log_2 P_e$$

# Fano Inequality

- Let $X$ and $X'$ be two independent identically distributed random variables with entropy $H(X)$. We have

$$P(X = X') = \sum_x p^2(x)\,.$$

  - The events $\{X = x, X' = x\}$ for $x \in \mathcal{X}$ are independent and $P(X = x, X' = x) = p^2(x)$.
- We have $P(X = X') \geq 2^{-H(X)}$ with equality if and only if $X$ has a uniform distribution.
  - Suppose that $X \sim p(x)$. By Jensen Inequality, we have $2^{E(\log_2 p(X))} \leq E\left(2^{\log_2 p(X)}\right)$.

$$2^{-H(X)} = 2^{\sum_x p(x) \log_2 p(x)} \leq \sum_x p(x) 2^{\log_2 p(x)} = \sum_x p^2(x)\,.$$

  - Exponential function $2^z$ is convex.

# Fano Inequality

- Let $X$ and $X'$ be two independent random variables with $X \sim p(x), X' \sim r(x)$, $x \in \mathcal{X}$. Then,

$$P(X = X') \geq 2^{-H(p)-D_r(p)}, P(X = X') \geq 2^{-H(r)-D_p(r)}$$

$$2^{-H(p)-D_r(p)} = 2^{\sum_x p(x)\log_2 p(x)+\sum_x p(x)\log_2 \frac{r(x)}{p(x)}} = 2^{\sum_x p(x)\log_2 r(x)} \leq \sum_x p(x)2^{\log_2 r(x)}$$

$$= \sum_x p(x)r(x) = P(X = X')$$